Software Requirements Specification (SRS) Summary

Project: Smart Irrigation Alert Dashboard
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1. Introduction

The **Smart Irrigation Alert Dashboard** is a web-based management and monitoring platform designed for modern precision agriculture. It continuously simulates and tracks soil moisture readings and ambient temperature across partitioned agricultural zones, triggering automated alerts when moisture drops below defined safety thresholds.

2. Functional Requirements (FR)

Requirement ID	Description
FR-01: Sensor Data Ingestion	The system shall provide REST APIs to ingest or simulate soil moisture (0–100%) and ambient temperature (-10°C to 60°C) readings for multiple named zones.
FR-02: Real-time Dashboard View	The web UI shall display a dynamic dashboard presenting all active sensor zones with their current moisture status, temperature, and last updated timestamp.
FR-03: Summary Metrics Cards	The UI shall compute and display global KPI summary indicators: Average Soil Moisture (%), Total Active Zones, and Total Critical Alerts.
FR-04: Automated Alert Engine	The system shall flag any zone with soil moisture $< 30\%$ as "Critical" and display a prominent warning banner and badge within 1 second of data arrival.
FR-05: Zone Search & Filtering	Users shall be able to filter the dashboard by zone name (text search) and by status category (All, Critical, Optimal, Saturated).
FR-06: Zone Detail Drill-Down	Clicking any zone card/row shall display historical readings, threshold parameters, and a manual irrigation trigger button.
FR-07: Zone Creation	Users shall be able to add new zones dynamically with custom crop types and initial soil readings.

3. Non-Functional Requirements (NFR)

NFR Category	Requirement Specification
Performance	Web dashboard page load time < 1.5 seconds; REST API response time < 200 ms.
Reliability & Availability	System shall achieve 99.9% uptime in local/container deployment; in-memory H2 database ensures zero external database dependency for the MVP.
Security	Input sanitization on all forms; non-root user execution in Docker container.
Portability	Multi-platform execution on Windows, Linux, and Docker runtime environments.
Maintainability	Clean Spring Boot MVC layering (Controller, Service, Repository, Model).

4. Use Case Diagram

```
flowchart LR
    Farmer((Farmer / Agronomist))
    Admin((System Admin))

    subgraph System["Smart Irrigation Alert System"]
        UC1([View Real-Time Dashboard])
        UC2([Filter & Search Zones])
        UC3([Inspect Zone Drill-Down])
        UC4([Receive Critical Moisture Alert])
        UC5([Trigger Manual Irrigation Simulation])
        UC6([Add New Sensor Zone])
        UC7([Deploy & Monitor Pipeline])
    end

    Farmer --> UC1
    Farmer --> UC2
    Farmer --> UC3
    Farmer --> UC4
    Farmer --> UC5

    Admin --> UC1
    Admin --> UC6
    Admin --> UC7
```